

Module End Assignment

“Data Cleaning, Analysis & Visualization with Excel”

Loyalty Level Data Analysis using Excel

1. Project Overview

The objective of this project is to analyze state wise loyalty level data using Microsoft Excel and generate meaningful business insights through data cleaning, transformation, analysis, and visualization.

This project demonstrates key Excel skills required for a Data Analyst role:

- Data Cleaning & Validation
- Formulas and Functions
- Pivot Tables & Pivot Charts
- Dashboard Creation

2. Business Problem

Management wants to understand the following business questions to drive data-driven decisions:

- Which State has more loyalty holders in overall and percentage wise.
- Which gender has more premium loyalty and Which gender has the least
- Age wise loyalty holders

3. Dataset Information

Dataset Name – Loyalty Level Dataset | Records – 500 accounts.

Column Name	Description
Member ID	Unique Loyalty ID
Member Name	Membership Holder Details
Age	Membership Holder Details
Gender	Membership Holder Details
City	Location of Member
State	Location of Member
Country	Location of Member
Loyalty Level	Categories of Membership

4. Data Cleaning Process

Steps	Action
Removing Duplicates	Data → Remove Duplicates
Handling Missing Values	Blank cells identified and corrected
Standardize Data	capitalization, Removing prefixes & Suffixes, Special Characters

5. Excel Functions Used

Formula	Purpose
Text.split	To remove Prefix & Suffix
Vlookup	Merging
Min	To find minimum % of Age group which holds the loyalty
Max	To find maximum % of Age group which holds the loyalty
Sum	To find the total number of Males & Females
IS	To find any blanks in the Age column

6. Pivot Table Analysis

Gender Wise Loyalty	
Male	238
Female	262
Overall	500

Insight – Overall Females hold more loyalty than Male

Age Wise Loyalty (%)	
Group - 1 (18 - 19 Yrs)	4%
Group - 2 (20 - 30 Yrs)	20%
Group - 3 (31 - 40 Yrs)	14%
Group - 4 (41 - 49 Yrs)	23%
Group - 5 (50 - 59 Yrs)	26%
Group - 6 (60 - 64 Yrs)	13%
Overall	100%

Insight – Group 5 (Aged between 50 – 59 yrs) were leading with highest percentage of loyalty holders and Group 1 (Aged between 18 – 19 yrs) were the lowest percentage of loyalty holders.

State Wise Loyalty (%)	
Minimum (Illinois)	1%
Maximum (Kansas)	5%

Insight – Overall 5% of loyalty holders were from Kansas and 1% of loyalty holders were from Illinois.

7. Visualizations Created

Chart Type	Purpose	Key Insight
Column Chart	Compare Gender wise Loyalty	Female has more loyatly level
Pie Chart	Age wise Comparison	Group 5 (50 - 59 yrs) hold more loyalty level
Bar Chart	Statewise Comparison in %	Kansas has 5% of Loyalty level holders
Line Chart	Statewise overall comparsion	Kansas holds 16 numbers of loyalty holders Including both male and female gender

8. Dashboard Components

- Pivot charts – Linked with live data
- Slicers – State wise, Gender wise and Loyalty level wise

9. Key Business Insights

- Female genders holds the highest number of Loyalty level holders.
- Group 5 (Aged 50 – 59 yrs) has the highest percentage of loyalty membership holders.
- Also, states like Kansas & Vermont holds the highest number of membership holders including both men and women.
- Overall Platinum pass holders are high.

10. Skills Demonstrated

Data Cleaning	Dash Board Design
Data Validation	Data Visualization
Conditional Formatting	Key Insights
Excel functions & Formulas	
Pivot tables & Charts	

11. Conclusion

This Excel Data Analyst project successfully transformed raw state wise loyalty holders data into actionable business insights. Using Excel's analytical capabilities — including pivot tables,

formulas, dashboards, and visualizations — management can now make informed decisions to improve revenue, profitability, and operational efficiency